

Translated

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Linear Diophantine Equation

A Linear Diophantine Equation (in two variables) is an equation of the general form:

$$ax + by = c$$

where a, b, c are given integers, and x, y are unknown integers.

In this article, we consider several classical problems on these equations:

- finding one solution
- finding all solutions
- finding the number of solutions and the solutions themselves in a given interval
- finding a solution with minimum value of $x + y$

The degenerate case

A degenerate case that need to be taken care of is when $a = b = 0$. It is easy to see that we either have no solutions or infinitely many solutions, depending on whether $c = 0$ or not. In the rest of this article, we will ignore this case.

Analytic solution

When $a \neq 0$ and $b \neq 0$, the equation $ax + by = c$ can be equivalently treated as either of the following:

$$\begin{aligned} ax &\equiv c \pmod{b} \\ by &\equiv c \pmod{a} \end{aligned}$$

Without loss of generality, assume that $b \neq 0$ and consider the first equation. When a and b are co-prime, the solution to it is given as

$$x \equiv ca^{-1} \pmod{b},$$

where a^{-1} is the [modular inverse](#) of a modulo b .